

FARBOD HAERI

✉ 4Farbodh@gmail.com | ☎ (916) 494-4394 | 🌐 farbodh.com | 🔗 LinkedIn | 🐙 GitHub | 📍 San Diego, CA

EDUCATION

University of California, San Diego

M.S. Mechanical Engineering

B.S. Mechanical Engineering, Specialization in Robotics & Controls

San Diego, CA

2026 – 2027

2024 – 2026

SKILLS

Controls: PID, LQR, lead lag, state space modeling, loop shaping, pole placement, notch filtering, sensor fusion, path following

Robotics & Vision: ROS 2, PX4, Pixhawk, Raspberry Pi, VESC, RTK GPS, LiDAR, OpenCV, YOLOv8, ONNX, DepthAI

Programming: Python, MATLAB, C++, JavaScript, Java, Flask, Git, Linux, Docker

CAD & Simulation: SolidWorks (Simulation, PDM), Fusion 360, Siemens NX, AutoCAD, ANSYS FEA

Manufacturing: CNC mill, manual lathe, waterjet, FDM 3D printing, soldering

Test & Measurement: thrust stand, motion capture, fatigue testing, optical metrology

WORK EXPERIENCE

Student Engineer Intern

County of Sacramento

May 2024 – Aug 2024

Sacramento, CA

- Benchmarked and scored energy efficiency across 120+ county buildings.
- Rebuilt utility histories for 120+ buildings into clean baselines and flagged the low performers for audit.

Materials Manager

Apple

Jun 2022 – Apr 2024

Elk Grove, CA

- Planned and kitted inventory across 500+ SKUs for new product builds.
- Flashed and tested production units and escalated supply risks before they reached the line.

RESEARCH EXPERIENCE

Graduate Researcher, UAV Controls & Robotics

Prof. Thomas Bewley · UC San Diego

Jan 2026 – Present

San Diego, CA

- Designed and built a fully actuated hexacopter, 674 g fully mounted. Fixed cant rotors let it translate without tilting.
- Designed the airframe in SolidWorks and printed it as a single monocoque.
- Picked the propulsion from thrust stand tests on ducted and open propellers.
- Deriving the 6-DoF controller. Gains tuned in MATLAB, checked against motion capture.

Undergraduate Researcher, Advanced Materials & Metrology

Prof. Maziar Ghazinejad · UC San Diego

Aug 2025 – Dec 2025

San Diego, CA

- Wrote MATLAB pipelines that measure defect fraction and surface roughness of 3D printed steel from metrology images.
- Designed 10+ components, machined test fixtures on the CNC mill and waterjet, and ran fatigue tests.

PROJECTS

Grove: Autonomous Environmental Survey Rover | Team Lead, UC San Diego

Aug 2026 – Sep 2026

- Designed and built a ROS 2 autonomous vehicle on a Raspberry Pi 5 with a VESC, OAK-D camera and RTK GPS.
- Wrote the lane detection and got the car driving a saved route on its own.
- Trained a detector for the onboard AI accelerator (0.735 mAP50, 41.8 fps) and tied each detection to its GPS position.
- Built the web interface that runs the car, shows the live feed, and writes an LLM survey report at the end of the lap.

Automated Cognitive-Training System for Mice | Senior Design, Jared Young Lab, UC San Diego

Jan 2026 – Jun 2026

- Built the operator dashboard the lab uses to launch sessions and watch each mouse's reaction time and accuracy live.
- Wrote the Python supervisor: a training state machine, a per-mouse scheduler and the MQTT link to the touchscreen rigs.